Problem 16.10

Chuck needs to purchase an item in 10 years. The item costs \$200 today, but its price increases by 4% per year. To finance the purchase, Chuck deposits \$20 into an account at the beginning of each year for 6 years. He deposits an additional X at the beginning of years 4, 5, and 6 to meet his goal. The annual effective interest rate is 10%. Calculate X.

Solution

Let the annual effective interest rate be

$$i = 0.10,$$

so

$$v = \frac{1}{1.10}.$$

The cost of the item in 10 years is

$$200(1.04)^{10}.$$

We value all deposits and the future purchase price at time 0.

The deposits of \$20 are made at the beginning of each year for 6 years, so their present value is

$$20\ddot{a}_{\overline{6}|}.$$

The additional deposits of X are made at the beginning of years 4, 5, and 6, which correspond to times 3, 4, and 5. Their present value is

$$X(v^3 + v^4 + v^5).$$

The present value of the required purchase price is

$$200(1.04)^{10}v^{10}.$$

Thus,

$$20\ddot{a}_{\overline{6}|} + X(v^3 + v^4 + v^5) = 200(1.04)^{10}v^{10}.$$

Solving for X gives

$$X = \frac{200(1.04)^{10}v^{10} - 20\ddot{a}_{\overline{6}|}}{v^3 + v^4 + v^5}.$$

Using

$$\ddot{a}_{\overline{n}|} = (1+i)\left(\frac{1-(1+i)^{-n}}{i}\right),$$

we get

$$\begin{aligned} 20\ddot{a}_{\overline{6}|} &= 20(1.10)\left(\frac{1-(1.10)^{-6}}{0.10}\right) \\ &\approx 95.8157, \end{aligned}$$

and

$$200(1.04)^{10}(1.10)^{-10} \approx 114.1397.$$

Therefore,

$$\begin{aligned} X &= \frac{114.1397 - 95.8157}{(1.10)^{-3} + (1.10)^{-4} + (1.10)^{-5}} \\ &\approx 8.9157. \end{aligned}$$

Hence,

$$\boxed{X \approx \$8.92}.$$